AI Model Inventory

Internal use case / Settlement failure early-warning model / Review snapshot: 09 Nov 2026

INHERENT RISK
HIGH
Financial operations exposure

RESIDUAL RISK
MEDIUM
After documented controls

DEPLOYMENT
LIVE
Production since 15 Apr 2026

01 / INVENTORY

Ownership, scope and technology

MODEL INVENTORY ID	INT-SET-042
APPLICATION	Settlement Failure Early Warning
USE CASE	Predict and prioritize likely US equity settlement failures
BUSINESS UNIT	Post-Trade Risk Analytics
BUSINESS OWNER	Maya Chen, Director of Settlement Risk
AI/MODEL TYPE	LightGBM gradient boosting + rule guardrails
MODEL PROVIDER	Internally developed / LightGBM v4.5.0
MODEL VERSION	r2.3.1
PRODUCTION DATE	15 April 2026
DEPLOYMENT STATUS	Production
DATA CLASSIFICATION	Confidential market participant transaction data
DATA SOURCES	Synthetic settlement events; exception queues; counterparty flags
FINANCIAL CRITICALITY	HIGH - impacts exception prioritization
DECISION AUTONOMY	Advisory output only; no automatic settlement actions
HUMAN OVERSIGHT	Operations analyst signs off on interventions
VALIDATION OUTCOME	PASS WITH OBSERVATIONS
VALIDATION DATE	02 October 2026
RISK RATINGS	Inherent HIGH / Residual MEDIUM
MONITORING FREQUENCY	Daily KPI dashboards; monthly drift review
NEXT INDEPENDENT REVIEW	15 January 2027

02 / EXECUTIVE SUMMARY

Why this model matters

The model ranks settlement exceptions by the likelihood of failure, allowing operations teams to investigate high-risk items earlier. It does not approve, modify, or settle transactions.

Key governance control

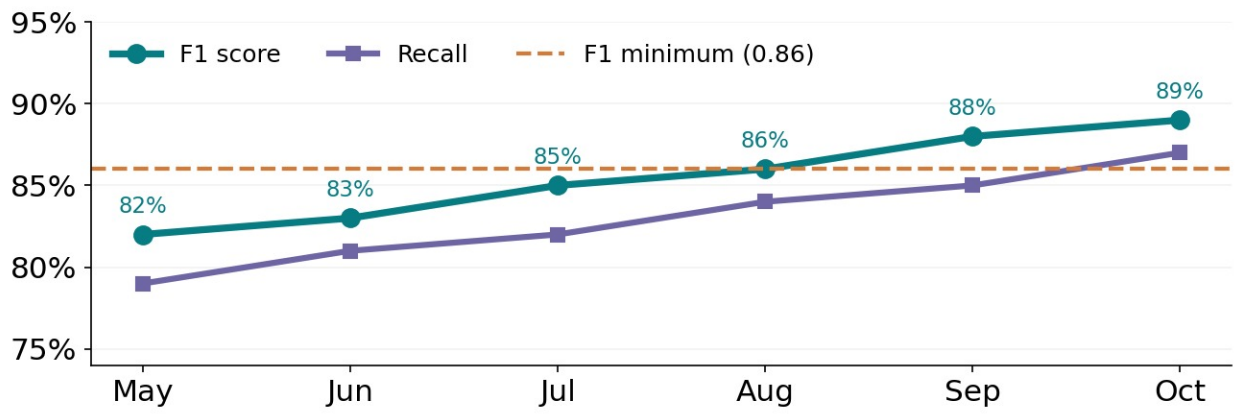
All recommended interventions require a named human approver; the manual exception queue remains available as a fallback.

Model Performance & Validation

The series and matrix below are raster graphics; underlying values are not PDF text objects.

03 / PERFORMANCE CHART

F1 score and recall by monitoring month



04 / CONFUSION MATRIX

October 2026 validation sample

	PREDICTED FAILURE: Yes	No
Actual failure	174	26
Actual success	17	783

n=1,000 records | Precision 91.1% | Recall 87.0%

05 / VALIDATION DECISION

Independent review summary

PRECISION
91.1%

RECALL
87.0%

F1 SCORE
89.0%

ACCEPTANCE
F1 >= 86%

OUTCOME
PASS WITH OBSERVATIONS

06 / REVIEW OBSERVATIONS

Open validation recommendations

V-11 Recalibrate threshold after quarterly data regime review.

V-12 Expand scenario tests for late-day high-volatility settlements.

V-13 Evidence the monthly drift sign-off in the model inventory.

Monitoring & Incident History

Visual risk heatmap and incident log / Snapshot: October 2026

07 / RISK HEATMAP

Inherent risk compared with strength of controls

IMPACT AREA	EXPOSURE	CONTROLS	RESIDUAL RISK
Settlement timeliness	HIGH	STRONG	MEDIUM
Confidentiality	HIGH	STRONG	MEDIUM
Model drift	MEDIUM	MODERATE	MEDIUM
Explainability	MEDIUM	STRONG	LOW
Operational fallback	HIGH	STRONG	LOW

Risk ratings are fictional and illustrative; they are not DTCC policy thresholds.

08 / INCIDENTS

Reported production events in trailing 12 months

EVENT ID	DATE	SEVERITY	OBSERVED ISSUE	STATUS
INC-0261	21 Aug 2026	Minor	Temporary input lag	Resolved / 4h
INC-0188	07 Jun 2026	Moderate	Recall below alert threshold	Resolved / 2d

Incidents in past 12 months: 2 | Both closed | No automated actions occurred.

09 / MONITORING CADENCE

Watchpoints and escalation

Daily	Precision / recall alarms and data completeness checks
Weekly	Operations sampling and overrides review
Monthly	PSI drift assessment; signed evidence packet
Trigger	PSI > 0.20 or recall < 0.82 requires escalation

Controls, Fallback & Sign-off

Visual control sequence and approval matrix / All information is fictional

10 / HUMAN-IN-THE-LOOP WORKFLOW

No autonomous client or settlement instructions

1. **Inbound exception** — Ingest trade exceptions
2. **Risk scoring** — Model ranks likely fails
3. **Analyst review** — Human verifies evidence
4. **Authorized action** — Approved workflow only

Fail-safe: manual exception queue takes over if scoring or integration is unavailable.

11 / CONTROL REGISTER

Documented mitigation and accountable party

CONTROL ID	MITIGATION	OWNER	STATE
C-01	Human authorization	Operations Lead	ACTIVE
C-02	Input data validation	Data Engineering	ACTIVE
C-03	Monthly PSI review	Model Risk	ACTIVE
C-04	Manual fallback drill	Operations	TESTED 18 OCT
C-05	Vendor access review	Information Security	N/A

12 / GOVERNANCE RECORD

Approval, retention and next review

APPROVING COMMITTEE
Model Risk Committee

RISK ACCEPTANCE OWNER
SVP, Post-Trade Operations

EVIDENCE RETENTION
13 months in encrypted audit storage

POLICY REFERENCE
AI-MRM-07 (fictional sample reference)

NEXT SCHEDULED REVIEW
15 January 2027